Status	Finished
Started	Monday, 19 October 2026, 5:21 PM
Completed	Monday, 19 October 2026, 5:23 PM
Duration	2 mins 27 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It does not require recursion
b. It is based on the divide-and-conquer approach
c. It sorts data using a single pass
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **2**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Searching
b. Sorting
c. Merging
d. Rearranging

The correct answer is: Sorting

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Binary Search
- b. Linear Search
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Merge Sort
- b. Bubble Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Bubble Sort
- b. Quick Sort
- c. Merge Sort
- d. Insertion Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort cannot handle duplicate elements
b. Bubble Sort does not guarantee sorted order
c. Bubble Sort is difficult to implement
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. The parts are merged without sorting
b. The parts are ignored
c. Each part is sorted independently
d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently

Question **8**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It complicates the selection of items
b. It makes the data unsorted
c. It makes it easier to select items based on their relationship to the rest of the items
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **9**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It only works with integer arrays
b. It requires external libraries
c. It is less efficient than custom sorting algorithms
d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Combining sorted sublists
c. Sorting the entire list sequentially
d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Bubble Sort
b. Quick Sort
c. Heap Sort
d. Merge Sort

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

a. Selection sort
b. Merge sort
c. Quick sort
d. Heap sort

The correct answer is:
Merge sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It only works on small datasets
c. It allows for efficient parallel processing and sorting of data
d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **14**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Makes code execution slower
b. Makes data harder to manage
c. Provides a basis for other algorithms to work efficiently
d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **15**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the largest element to its correct position
- b. Sorting the entire list in one pass
- c. Bubbling up the smallest element
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position